

Question Paper

Exam Date & Time: 18-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Computer Vision-III [CSE_4031]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) A drone captures aerial images of a construction site to monitor progress and detect structural cracks. (5)
- A) Illustrate how digital image processing steps help to extract structural details and detect possible defects.
- Course Outcome: "CO1" / Cognitive Level: "Understand"
- B) Examine how Power-Law Transformations are used to control image contrast in photographs captured under varying illumination conditions. (5)
- Course Outcome: "CO1" / Cognitive Level: "Analysis"
- 2) Illustrate how edge detection techniques help in obtaining important structural details from digital images. (5)
- A) Compare the working principles of the Laplacian and Canny edge detection methods
- Course Outcome: "CO2" / Cognitive Level: "Understand"
- B) Apply the concept of the Histogram of Oriented Gradients descriptor to identify and represent the shape and structure of objects within an image. Discuss with a suitable real time application (5)
- Course Outcome: "CO2" / Cognitive Level: "Application"
- 3) A drone captures rooftop images for inspection, but the photos appear misaligned and distorted due to the camera's tilt. Apply geometric transformations to adjust the tilted drone images and discuss in detail. (5)
- A)
- Course Outcome: "CO3" / Cognitive Level: "Application"
- B) In an AR game, players use their phone camera to find virtual coins, but lighting changes and camera shake cause feature mismatches, making the coins appear misplaced. Apply the RANSAC algorithm to detect and remove incorrect feature points, helping the game build a stable surface model and accurately place virtual coins on real surfaces. (5)
- Course Outcome: "CO3" / Cognitive Level: "Application"
- 4) Apply the learned motion estimation framework to detect abnormal motion patterns in video streams and discuss in detail. (5)
- A)

Course Outcome: "CO4" / Cognitive Level: "Application"

- B) In a video tracking application, a system needs to estimate how an object moves between consecutive frames. Utilize the Lucas-Kanade method to estimate the motion of objects across consecutive frames. (5)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 5) Develop an intruder detection system using face recognition for a secure facility where individuals may wear masks, glasses, or hats and explain the recognition pipeline to create a reliable method that ensures accurate identification of unauthorized persons under varying appearance conditions. (5)
- A)

Course Outcome: "CO5" / Cognitive Level: "Application"

- B) Apply a computer vision-based activity detection model to develop an elderly care monitoring system that observes indoor movements and detects signs of falls or distress. How would you apply these techniques to automatically identify normal vs. abnormal activities and trigger alerts in real time? (5)

Course Outcome: "CO5" / Cognitive Level: "Application"

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Event Name	Subject
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058-END SEMSTER-ODD-2025 CSE_4031:Computer Vision-III

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025 CSE_4031:Computer Vision-III

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	2	10(20%)
3	Application	7	35(70%)
4	Analysis	1	5(10%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		10	50 (100%)